

Snap Dataset — Full Data Intelligence Audit

For Investor Insight AI System Design

Audit Date: 2026-03-18 **Database:** Supabase Project `ojyxblegxpdgaqiscxpz` **Total Tables:** 61 | **Total Views:** 9 | **Total Functions:** 100+ **Migrations:** 229 SQL files

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1. Database Architecture Overview

Table Classification

Category	Tables	Purpose
Core Property Data	<code>properties</code> , <code>violations</code> , <code>clean_leads</code> , <code>property_contacts</code>	Main data for Investor Insight
Upload Pipeline	<code>upload_jobs</code> , <code>upload_staging</code> , <code>upload_history</code> , <code>staging_uploads</code>	CSV ingestion system

Category	Tables	Purpose
Jurisdiction/ FOIA	jurisdictions , counties , targets , foia_requests , foia_assignments , foia_profiles , foia_invites , foia_templates	Data acquisition workflow
User/Auth	profiles , user_profiles , user_roles , user_subscriptions , user_allowed_states , user_invitations , organizations	User management
Engagement	lead_activity , lead_lists , list_properties , saved_properties , call_logs	CRM / lead tracking
Credits/Billing	credit_ledger , credit_ledger_skiptrace , subscription_plans , subscription_usage , user_subscriptions	Monetization
Skip Trace	skiptrace_jobs , skiptrace_outcomes , skiptrace_bulk_items , skiptrace_bulk_runs , skiptrace_consent_log	Contact lookup
Communication	email_templates , sms_templates , email_analytics , email_preferences	Outreach
System	error_logs , system_logs , events , webhook_events , webhook_errors , export_logs , user_activity_log	Operational logging
Press/VA	press_accounts , press_rotation , rotation_alerts , va_credential_slots , credential_target_cooldown	FOIA operations
Geo/Spatial	spatial_ref_sys , census_places , geocoding_jobs	PostGIS / mapping
Misc	beta_waitlist , list_enrichment_waitlist , enrichment_jobs	Waitlists / enrichment

Key Views for Investor Insight

View	Purpose	Fields
v_hot_properties	Pre-filtered high-distress properties	address, city, state, distress_signals, escalated, snap_score, total_violations
v_jurisdiction_stats	Aggregated city-level enforcement data	avg_score, property_count, distressed_count, enforcement_profile
v_opportunity_funnel		opportunity_class, property_count, avg_score

View	Purpose	Fields
	Properties grouped by opportunity class	
v_property_contact_stats	Contact info completeness per property	contact_rows, emails_found, phones_found
v_user_credits	User credit balance	balance, user_id
mv_distinct_cities	Materialized: unique cities	city, state
mv_distinct_states	Materialized: unique states	state

2. Core Data Tables — Schema Breakdown

properties — Central Property Record (31 fields)

Field	Type	Nullable	Investor Insight Relevance
id	uuid	NO	Primary key
address	text	NO	Core identifier
city	text	NO	Location filter
state	text	NO	Location filter
zip	text	NO	Location filter
county	text	YES	Location filter
latitude	float	YES	Map display
longitude	float	YES	Map display
geom	geometry	YES	PostGIS spatial queries
snap_score	integer	YES	PRIMARY — 0-100 enforcement intensity score
snap_insight	text	YES	PRIMARY — AI/rule-based narrative summary
total_violations	integer	YES	HIGH — violation count
open_violations	integer	YES	HIGH — active enforcement count

Field	Type	Nullable	Investor Insight Relevance
violation_types	text[]	YES	HIGH — array of violation category strings
distress_signals	text[]	YES	HIGH — array of signal keywords
opportunity_class	text	YES	HIGH — "distressed" / "value_add" / "watch"
enforcement_type	text	NO (default "")	HIGH — "water_shutoff" or standard
repeat_offender	boolean	YES	HIGH — 3+ violations flag
multi_department	boolean	YES	HIGH — 2+ enforcement categories
escalated	boolean	YES	HIGH — condemned/legal/court status
avg_days_open	float	YES	MEDIUM — average enforcement duration
oldest_violation_date	date	YES	MEDIUM — enforcement history start
newest_violation_date	date	YES	MEDIUM — most recent activity
last_enforcement_date	date	YES	MEDIUM — last enforcement action
last_analyzed_at	timestamp	YES	Freshness indicator
scope	text	YES	"city" or "county" upload scope
jurisdiction_id	uuid	YES	FK → jurisdictions
photo_url	text	YES	Property image
created_at	timestamp	YES	Record creation
updated_at	timestamp	YES	Last modification

violations — Individual Violation Records (16 fields)

Field	Type	Nullable	Investor Insight Relevance
id	uuid	NO	Primary key
property_id	uuid	YES	FK → properties
violation_type	text	NO	PRIMARY — violation category label
status	text	NO	PRIMARY — "Open" / "Closed" / escalated statuses
case_id	text	YES	Municipal case reference

Field	Type	Nullable	Investor Insight Relevance
description	text	YES	Normalized description
raw_description	text	YES	HIGH — original text from source CSV
days_open	integer	YES	HIGH — duration metric
opened_date	date	YES	When violation was filed
closed_at	date	YES	When violation was resolved
last_updated	date	YES	Most recent status change
first_seen_at	timestamp	YES	When Snap first ingested
last_seen_at	timestamp	YES	Most recent Snap observation
previous_status	text	YES	Status before current
status_changed_at	timestamp	YES	When status last changed
created_at	timestamp	YES	Record creation

clean_leads — Cleaned Lead Records (15 fields)

Field	Type	Nullable	Notes
id	uuid	NO	Primary key
address	text	NO	Cleaned address
city	text	NO	City
state	text	NO	State code
zip	text	YES	Zip code
violation_type	text	YES	Violation category
violation_description	text	YES	Description text
snap_score	integer	YES	Score at time of cleaning
snap_insight	text	YES	Insight at time of cleaning
property_id	uuid	YES	FK → properties
county_id	uuid	YES	FK → counties

Field	Type	Nullable	Notes
<code>opened_date</code>	date	YES	Violation date
<code>last_updated</code>	date	YES	Last update
<code>created_by</code>	uuid	YES	User who created
<code>created_at</code>	timestamp	YES	Record creation

`property_contacts` — Skip Trace Results (9 fields)

Field	Type	Nullable	Notes
<code>id</code>	uuid	NO	Primary key
<code>property_id</code>	uuid	NO	FK → properties
<code>name</code>	text	YES	Contact name
<code>email</code>	text	YES	Contact email
<code>phone</code>	text	YES	Contact phone
<code>source</code>	text	YES	Data provider
<code>raw_payload</code>	jsonb	YES	Full vendor response
<code>created_by</code>	uuid	NO	User who requested
<code>created_at</code>	timestamp	NO	Record creation

`jurisdictions` — City Enforcement Profiles (9 fields)

Field	Type	Nullable	Notes
<code>id</code>	uuid	NO	Primary key
<code>name</code>	text	NO	Display name
<code>city</code>	text	NO	City name
<code>state</code>	text	NO	State code
<code>county</code>	text	YES	County name
<code>enforcement_profile</code>	jsonb	YES	HIGH — AI-generated enforcement behavior profile

Field	Type	Nullable	Notes
ai_summary	text	YES	AI summary of jurisdiction
default_zip_range	text	YES	Default zip code range
created_at	timestamp	NO	Record creation

3. Violation Type Analysis

Current Violation Classification System

The system classifies violations using a **keyword-scan approach** on `violation_type` + `raw_description` combined text. Here's the complete classification:

HIGH Priority Categories

Category	Keywords Scanned	Priority
Structural	collapse, unsafe structure, condemned, foundation failure, imminent danger	HIGH
Fire	fire damage, burnt, smoke damage, charred, fire, burn, smoke	HIGH
Utility	no utilities, utility disconnect, no water, no electric, water disconnect, water shutoff	HIGH

MEDIUM Priority Categories

Category	Keywords Scanned	Priority
Structural	roof leak, structural damage, foundation crack, major repair, structure, foundation, roof, wall	MEDIUM
Vacancy	vacant, abandon, unoccup, boarded	MEDIUM
Safety	unsafe, hazard, danger, health	MEDIUM
Utility	plumbing, electrical, sewage, hvac	MEDIUM
Zoning	zoning, zone violation, land use, code enforcement, unpermitted, without permit, permit violation, illegal construction	MEDIUM
Maintenance	property maintenance, property inspection, code compliance, nuisance	MEDIUM

Category	Keywords Scanned	Priority
General Enforcement	(any open violation not matching above)	MEDIUM

LOW Priority Categories

Category	Keywords Scanned	Priority
Exterior	paint, siding, fence, grass, weeds, debris, window, door, screen, gutter, exterior, facade	LOW
Other	(any closed violation not matching above)	LOW

Category Groupings Used in UI Filtering (`fn_properties_by_category`)

UI Category	Keyword Array
<code>exterior</code>	Exterior
<code>structural</code>	Structural
<code>safety</code>	Safety, Fire
<code>zoning</code>	Zoning
<code>maintenance</code>	Rubbish, Grass, Trash, Debris, Weed, Dumping, Waste, Snow
<code>interior</code>	Interior, Plumbing, HVAC, Furnace, 305.3, 305.6, 605.3, 403., 504., 506., 605.
<code>vacancy</code>	Vacancy, Vacant
<code>water_disconnection</code>	enforcement_type = 'water_shutoff'
<code>other</code>	Unknown, Other, Complaint

Upload-Time Violation Type Normalization (`normalizeViolationType`)

Before storing in the `violations` table, the raw CSV violation type is normalized:

Input Pattern (case-insensitive)	Normalized Output
fire, burn, smoke	<code>Fire</code>
unsafe, hazard, danger, safety	<code>Safety</code>
structur, foundation, roof, wall, collapse	<code>Structural</code>

Input Pattern (case-insensitive)	Normalized Output
electric, plumb, water, gas, sewage, utility	Utility
exterior, facade, siding, paint, window, door, grass, weed, overgrown, fence, yard	Exterior
(no match)	Original text trimmed to 50 chars, or Unknown

Upload-Time Status Normalization (`normalizeStatus`)

Input Pattern (case-insensitive)	Normalized Output
open, pending, active, in progress, new, referred, board, hearing	Open
closed, resolved, complete, complied, dismissed, abated	Closed
(no match)	Unknown

Potential Edge Cases in Violation Types

- 1. **Violation types ARE partially normalized at upload time** via `normalizeViolationType()` , but only 5 categories are mapped — everything else keeps its raw text (trimmed to 50 chars)
 - 2. **Classification happens at scoring time** — the `classifyViolation()` function runs keyword matching on the combined `violation_type + raw_description` text
 - 3. **Many violations may fall to "Other/General Enforcement"** if the source CSV uses city-specific codes (e.g., "305.3", "ICC 101.1", "CE-2024-xxxxx")
 - 4. **Description text is the richest source** but is optional and inconsistently provided across jurisdictions
 - 5. **Municipal code references** (e.g., "305.3", "605.3") are hardcoded in the interior category — not all code references are covered
 - 6. **No formal taxonomy** — violation types are free-text from CSV sources, meaning the AI must handle arbitrary strings
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4. Data Completeness Audit

properties Table — Field Completeness Assessment

Field	Required	Likely Fill Rate	Risk Level	Notes
address	YES	~100%	LOW	Required on insert
city	YES	~100%	LOW	Required; extracted from address if missing
state	YES	~100%	LOW	Required; falls back to job-level state
zip	YES	~95%	MEDIUM	Required on insert but can be empty string; backfill functions exist
county	NO	~30-50%	HIGH	Only populated on county-scope uploads
snap_score	NO	~80-90%	MEDIUM	Null until generate-insights runs; backfill available
snap_insight	NO	~80-90%	MEDIUM	Generated alongside snap_score
total_violations	NO	~80-90%	MEDIUM	Computed during scoring
open_violations	NO	~80-90%	MEDIUM	Computed during scoring
violation_types	NO	~80-90%	MEDIUM	Array; computed during scoring
distress_signals	NO	~80-90%	MEDIUM	Array; computed during scoring
opportunity_class	NO	~80-90%	MEDIUM	Computed during scoring
enforcement_type	YES (default "")	~100%	LOW	Default empty string
repeat_offender	NO	~80-90%	MEDIUM	Boolean; computed (3+ violations)
multi_department	NO	~80-90%	MEDIUM	Boolean; computed (2+ categories)
escalated	NO	~80-90%	MEDIUM	Boolean; computed from status keywords
latitude	NO	~60-70%	HIGH	Requires geocoding; batch function exists

Field	Required	Likely Fill Rate	Risk Level	Notes
longitude	NO	~60-70%	HIGH	Requires geocoding; batch function exists
avg_days_open	NO	~80-90%	MEDIUM	Computed during scoring
oldest_violation_date	NO	~70-80%	MEDIUM	Depends on violation dates
newest_violation_date	NO	~70-80%	MEDIUM	Depends on violation dates
jurisdiction_id	NO	~50-70%	MEDIUM	Only linked if jurisdiction exists
photo_url	NO	~5-10%	HIGH	Rarely populated

violations Table — Field Completeness Assessment

Field	Required	Likely Fill Rate	Risk Level	Notes
property_id	NO	~99%	LOW	FK to properties; almost always linked
violation_type	YES	~100%	LOW	Required; but may be generic (e.g., "Complaint")
status	YES	~100%	LOW	Required; typically "Open" or "Closed"
case_id	NO	~60-80%	MEDIUM	Not all jurisdictions provide case numbers
description	NO	~30-50%	HIGH	Normalized description; sparsely filled
raw_description	NO	~40-60%	HIGH	Original CSV text; most valuable for AI
days_open	NO	~50-70%	HIGH	Computed from dates; null if no dates available
opened_date	NO	~60-80%	MEDIUM	Key date; not all CSVs provide it
closed_at	NO	~20-40%	HIGH	Only for resolved violations
last_updated	NO	~40-60%	HIGH	Inconsistently provided
first_seen_at	NO	~50-60%	MEDIUM	Upload tracking
last_seen_at	NO	~50-60%	MEDIUM	Upload tracking
previous_status	NO	~10-20%	HIGH	Only populated on status changes

Field	Required	Likely Fill Rate	Risk Level	Notes
<code>status_changed_at</code>	NO	~10-20%	HIGH	Only populated on status changes

Data Quality Variation by Source

Factor	Impact on Data Quality
City vs County scope	City-scoped uploads have more complete city fields; county-scoped may have null city
FOIA response format	CSV, PDF, image — only CSVs produce structured data; PDFs/images need manual processing
Jurisdiction compliance	Some cities provide rich violation descriptions; others provide only codes
State coverage	Currently focused on CA, NV cities (known city list in upload parser)
Date formats	Parser handles ISO (YYYY-MM-DD) and US (MM/DD/YYYY) formats
Redacted records	FOIA responses may redact owner info, addresses

5. Field Structure & Naming Consistency

Cross-Table Field Name Inconsistencies

Concept	<code>properties</code>	<code>violations</code>	<code>clean_leads</code>	<code>upload_staging</code>
Violation type	<code>violation_types</code> (array)	<code>violation_type</code> (text)	<code>violation_type</code> (text)	<code>violation</code> (text)
Description	N/A	<code>description</code> + <code>raw_description</code>	<code>violation_description</code>	<code>raw_description</code>
Date opened	<code>oldest_violation_date</code>	<code>opened_date</code>	<code>opened_date</code>	<code>opened_date</code>
Date closed	N/A	<code>closed_at</code>	N/A	<code>last_updated</code>
Status	N/A	<code>status</code>	N/A	<code>status</code>

Concept	properties	violations	clean_leads	upload_staging
Location	city , state , zip , county	N/A (via property FK)	city , state , zip	city , state , zip
Case ID	N/A	case_id	N/A	case_id
Score	snap_score	N/A	snap_score	N/A
Insight	snap_insight	N/A	snap_insight	N/A

CSV Column Name Mappings (from process-upload)

The upload parser handles these CSV column name variations:

Data Field	Accepted CSV Column Names
Case ID	case_id , case/file id , file # , file_number , id , file number
Address	address , location , property_address , property address
Violation Type	category , violation , type , violation_type , violation type , violation_category
Open Date	opened_date , open_date , open date , date , date_opened
Close Date	close_date , close date , closed_date , date_closed
Description	description , violation_description , notes , comments
City	city
State	state
Zip	zip , zipcode , zip code
Status	status (defaults to "Open" if missing)

Issues for AI Prompt Design

1. **violation** in staging vs **violation_type** in violations — same concept, different field names
2. **description** vs **raw_description** vs **violation_description** — three different fields across tables for the same concept

3. **closed_at** vs **close_date** vs **last_updated** — close date stored differently across tables
4. **Status field overloading** — **upload_jobs.status** = job status (QUEUED/PARSING/PROCESSING/COMPLETE/FAILED) vs **violations.status** = violation status (Open/Closed/Board/Legal/Court)
5. **enforcement_type** **default empty string** — not null, but empty string "" when no special enforcement, "water_shutoff" when water disconnected

6. Text / Description Analysis

Violation Description Patterns

Based on the code analysis, violation descriptions contain:

Severity Signal Keywords (from classifyViolation)

Severity	Keywords	Example Descriptions
CRITICAL	collapse, unsafe structure, condemned, foundation failure, imminent danger, condemnation	"Structure condemned - unsafe for occupancy"
HIGH	fire damage, burnt, smoke damage, no water, water disconnect, water shutoff, no utilities	"Water service disconnected per utility authority"
MEDIUM	roof leak, structural damage, foundation crack, vacant, abandoned, boarded, unsafe, hazard, zoning, unpermitted	"Vacant property with boarded windows and overgrown vegetation"
LOW	paint, siding, fence, grass, weeds, debris, window, door, gutter	"Overgrown weeds and debris in front yard"

Escalation Status Keywords (from aggregatePropertyIntelligence)

Status	Interpretation
board	Referred to administrative board hearing
legal	Referred for legal enforcement
court	Municipal court proceedings
condemned	Property condemned as unsafe

Status	Interpretation
prosecution	Criminal prosecution initiated

Water Shutoff Detection Keywords

Keyword	Source
water shutoff	violation_type or raw_description
water disconnect	violation_type or raw_description
no water	violation_type or raw_description
water termination	violation_type or raw_description
water service disconnect	violation_type or raw_description

Address Validation Filters

The upload parser rejects addresses containing: - Street number-less text - Case number prefixes (e.g., "CE-2026-04026") - ATTN: prefixes - Concatenated status fields - AI/narrative text ("nextdoor", "posting says") - Complaint narratives ("complainant", "neighbor", "property owner") - OCR garbage text - Legal land descriptions (SEC, LOT, PTN) - Sentence structures

7. Frequency & Pattern Insights

Snap Score Distribution & Interpretation

Score Range	Activity Class	Opportunity Class	Interpretation
70-100	Critical	distressed	Active multi-vector enforcement; highest investor interest
40-69	Elevated	value_add	Significant enforcement activity; value-add potential
0-39	Monitoring	watch	Minor or resolved enforcement; monitoring only

Scoring Factors (from calculateEnforcementIntensity)

Factor	Max Points	Condition
Duration	30	3 pts per month open (capped at 30)
High-priority violations	60	40 base + 10 per additional (capped at +20)
Medium-priority violations	30	15 per violation (capped at 30)
Repeat activity	30	5 pts (2+), 15 pts (3+), 25 pts (5+), 30 pts (10+)
Open violation volume	70	Progressive: 10 (3+), 20 (5+), 30 (10+), 40 (20+), 50 (50+), 60 (100+), 70 (200+)
Multi-category	25	15 (2+ categories), 25 (3+ categories)
Escalation	30	15 (board), 25 (legal/court), 30 (condemned/prosecution)
Vacancy	25	Vacancy/abandonment citation detected
Recency	40	40 (7 days), 20 (30 days)
Water shutoff	55	40-55 depending on concurrent violations
Score cap: all closed	-cap	Max 20 (normal) or 35 (escalated) if no open violations

Distress Signals Array Values

Signal String	Meaning
extended_enforcement	Open violations > 180 days
fire_citation	Fire safety violations detected
structural_citation	Structural violations detected
recurring_enforcement	3+ total violations
multiple_citations	2+ total violations
extreme_enforcement_load	200+ open violations
massive_enforcement_load	50-199 open violations
high_violation_volume	10-49 open violations

Signal String	Meaning
<code>active_enforcement_load</code>	3-9 open violations
<code>coordinated_enforcement</code>	3+ enforcement categories
<code>multi_department</code>	2+ enforcement categories
<code>enforcement_escalation</code>	Condemned, legal, court, or board status
<code>vacancy_citation</code>	Vacancy/abandonment detected
<code>recent_activity</code>	Activity within 7 days
<code>current_enforcement</code>	Activity within 30 days
<code>water_shutoff_enforcement</code>	Water service disconnected
<code>maximum_enforcement_pressure</code>	Water shutoff + open code violations + repeat offender + recent activity
<code>active_enforcement_current</code>	Water shutoff + recent activity
<code>compounding_enforcement</code>	Water shutoff + open code violations
<code>direct_municipal_action</code>	Water shutoff only
<code>utility_enforcement</code>	Non-water utility violations

Multi-Violation Property Patterns

Pattern	Detection	Investor Relevance
Single violation	<code>total_violations = 1</code>	Low distress indicator
Multiple violations	<code>total_violations >= 2</code>	Moderate distress
Repeat offender	<code>total_violations >= 3</code> (sets <code>repeat_offender = true</code>)	Strong distress signal
Multi-department	2+ unique enforcement categories	Cross-agency involvement
Escalated	Status contains board/legal/court/condemned/ prosecution	Severe enforcement action
Water shutoff + violations	<code>enforcement_type = 'water_shutoff'</code> AND open violations	Maximum distress indicator

8. Data Outliers & Edge Cases

Known Edge Cases That Could Break AI Interpretation

Edge Case	Description	Impact	Recommendation
Empty violation_type	Some CSVs have blank type columns	Falls to "Other" category	AI must handle empty/generic types
Municipal code numbers as types	"305.3", "605.3", "ICC 101.1"	Only partially mapped in category filter	Build code-to-category lookup table
Status variations	"OPEN", "Open", "open", "Active", "Investigating", "CLOSED", "Resolved"	Status normalization happens at scoring	AI prompt must normalize status strings
Future dates	Dates parsed as future (parser rejects > 30 days out but edge cases exist)	May show negative days_open	Clamp to current date
Null days_open with no dates	Violations with no opened_date, last_updated, or close_date	days_open defaults to 0	Treat as "unknown duration"
Concatenated CSV fields	Addresses with embedded status ("816 E 2ND CLOSED 1/21/2026")	Parser filters these	Some may slip through
OCR-sourced data	PDFs converted to text; garbage characters	Parser has OCR filters	AI must handle garbled text
Duplicate properties	Same address uploaded from multiple CSV files	Deduplication by address match	May have inflated violation counts
County-scope nulls	County-level uploads have null city	Properties without city context	AI must handle missing city
Empty string enforcement_type	Default "" not null	Comparison must use empty string check	Use <code>enforcement_type = 'water_shutoff'</code> not null check
Stale scores	Properties not re-scored after new violations uploaded	Score/insight may be outdated	Check <code>last_analyzed_at</code> freshness

Edge Case	Description	Impact	Recommendation
Zero-violation properties	Properties created but violations not yet processed	snap_score = null, empty arrays	AI should state "no violations on file"
Very high violation counts	Properties with 100-200+ open violations	Score scaling handles up to 200+	AI must handle extreme counts gracefully
Escaped/multi-line descriptions	Raw CSV text with newlines, quotes	Papaparse handles; stored with spaces	AI should expect cleaned text

Invalid Address Patterns Filtered at Upload

Pattern	Examples	Filter Reason
No street number	"Main Street" without number	no_street_number
Case number prefix	"CE-2026-04026 CE - INOPERATIVE..."	case_number_prefix
ATTN prefix	"ATTN: Building Department"	attn_prefix
Concatenated status	"816 E 2ND CLOSED 1/21/2026"	concatenated_status
Narrative text	"Neighbor called to report..."	looks_like_description
Legal descriptions	"SEC 12 LOT 3 LYG..."	legal_description
OCR garbage	Multiple pipes, random chars	ocr_garbage
Too long (>100 chars)	Full paragraphs	too_long
Too short (<5 chars)	"N/A"	too_short

9. Investor-Relevant Signals

Fields That Signal Motivated Sellers / Financial Distress

Signal	Fields Used	Weight	Interpretation
Water Shutoff	enforcement_type = 'water_shutoff'	HIGHEST	Direct utility disconnection = severe financial distress or vacancy

Signal	Fields Used	Weight	Interpretation
Condemned/ Unsafe	<code>escalated = true</code> + description keywords	VERY HIGH	Property deemed uninhabitable; owner under legal obligation
High Snap Score (70+)	<code>snap_score >= 70</code>	HIGH	Multi-factor enforcement pressure indicates distressed owner
Repeat Offender	<code>repeat_offender = true</code> (3+ violations)	HIGH	Pattern of non-compliance suggests inability/unwillingness to maintain
Multi-Department	<code>multi_department = true</code>	HIGH	Multiple agencies involved; coordinated enforcement pressure
Extended Enforcement	<code>avg_days_open > 180</code>	HIGH	Long-standing unresolved issues; deferred maintenance
Recent Activity	<code>newest_violation_date</code> within 30 days	MEDIUM	Active enforcement; time-sensitive opportunity
Multiple Open Violations	<code>open_violations >= 5</code>	HIGH	Active enforcement load creates motivation
Vacancy Citation	<code>vacancy_citation</code> in <code>distress_signals</code>	MEDIUM	Property may be vacant/abandoned; owner may be motivated
Fire Damage	<code>fire_citation</code> in <code>distress_signals</code>	HIGH	Major damage; potential insurance/financial distress
Structural Issues	<code>structural_citation</code> in <code>distress_signals</code>	HIGH	Major repair costs may exceed property value
Legal/Court Referral	<code>enforcement_escalation</code> in <code>distress_signals</code>	HIGH	Legal pressure creates urgency

Opportunity Classification Matrix

Class	Score Range	Signals Required	Investor Action
Distressed	70-100	Critical activity class; water shutoff, condemned, 10+ violations, escalation	Immediate outreach; highest motivation
Value-Add	40-69	Elevated activity class; multiple violations, structural issues, vacancy	Strong opportunity; moderate urgency

Class	Score Range	Signals Required	Investor Action
Watch	0-39	Monitoring activity class; minor/resolved violations	Monitor for escalation

Urgency Indicators for Investor Insight AI

Urgency Level	Indicators	AI Response Should Include
IMMEDIATE	Water shutoff + open violations + recent activity	Fines accumulating, utility service disrupted, municipal deadlines
HIGH	Condemned/legal referral, 5+ open violations	Legal proceedings, board hearing dates, condemnation orders
MODERATE	3+ violations, structural issues, vacancy	Maintenance backlog, code compliance timeline, violation categories
LOW	1-2 resolved violations	Historical record, compliance history, no current action

10. AI Prompt System Recommendations

Data Fields the AI Prompt MUST Reference

For every property analyzed by Investor Insight AI, the prompt should include:

```

REQUIRED CONTEXT:
- address, city, state, zip
- snap_score (0-100)
- opportunity_class (distressed / value_add / watch)
- total_violations, open_violations
- violation_types[] (array of categories)
- distress_signals[] (array of signal keywords)
- enforcement_type ('' or 'water_shutoff')
- repeat_offender (boolean)
- multi_department (boolean)
- escalated (boolean)
- avg_days_open
- oldest_violation_date, newest_violation_date
- snap_insight (existing AI/rule-based summary)

```

OPTIONAL BUT VALUABLE:

- Individual violation records (violation_type, status, days_open, raw_description, opened_date)
- property_contacts (name, email, phone)
- jurisdiction_enforcement_profile (JSON)
- jurisdiction_ai_summary

Edge Cases the AI Prompt Must Handle

1. **No violations on file** → "No enforcement actions currently documented"
2. **All violations closed** → Focus on historical record, compliance timeline
3. **Null snap_score** → Property not yet scored; state this clearly
4. **Empty distress_signals_array** → No active distress indicators
5. **enforcement_type = "** → Standard code violation, NOT water shutoff
6. **Municipal code numbers** → Don't interpret as meaningful without context
7. **Very long raw_descriptions** → Truncated to 2000 chars; may end mid-sentence
8. **County-scope with null city** → Use county + state for location context
9. **Future dates** → Reject or flag as data error
10. **Zero days_open with open status** → Date data unavailable, not "just opened"
11. **Extremely high violation counts (100+)** → Likely systematic/portfolio enforcement
12. **Stale last_analyzed_at** → Score may not reflect latest violations

Recommended AI Prompt Structure for Investor Insight

SYSTEM: You are an Investor Insight AI that analyzes municipal code enforcement data to identify real estate investment opportunities.

RULES:

1. Always cite specific data: violation counts, categories, dates, scores
2. Clearly state the opportunity_class and what it means
3. Highlight the TOP distress signals and their investor implications
4. For water_shutoff properties, always mention utility disconnection
5. For escalated properties, specify the escalation type
6. Include a recommended action (contact owner, monitor, skip)
7. If data is incomplete (null scores, no descriptions), say so

PROPERTY DATA:

{Insert structured property + violations data}

OUTPUT: Provide a 3-section investor brief:

1. ENFORCEMENT SUMMARY (what's happening)

2. DISTRESS INDICATORS (why this matters to investors)
3. RECOMMENDED ACTION (what to do next)

Data Quality Improvements Needed Before Launch

Priority	Improvement	Impact
P0	Ensure all properties have snap_score (run bulk-rescore)	Eliminates null score edge case
P0	Backfill zip codes for all properties (run backfill-zips)	Completes location data
P1	Standardize violation_type values into canonical taxonomy	Enables accurate categorization
P1	Ensure raw_description is preserved for all violations	Richest data source for AI
P1	Geocode all properties with null lat/lng	Enables map features
P2	Add owner name / mailing address to properties table	Critical for investor outreach
P2	Track fine amounts from FOIA data (currently not in violations table)	Key urgency indicator
P2	Add enforcement_deadline field to violations	Time-sensitive opportunity indicator
P3	Build violation_type normalization lookup table	Reduce "Other" category
P3	Add property_value_estimate field	ROI calculation support

Summary Statistics

Metric	Value
Total database tables	61
Core property/violation tables	4 (properties, violations, clean_leads, property_contacts)
Database views	9
Edge functions	38+
SQL migrations	229

Metric	Value
Violation classification categories	10 (Structural, Fire, Utility, Vacancy, Safety, Zoning, Maintenance, Exterior, General Enforcement, Other)
Distress signal types	21 unique signal strings
CSV column mappings	10 data fields with 2-6 name variations each
Score range	0-100 (3 tiers: monitoring/elevated/critical)
Opportunity classes	3 (watch, value_add, distressed)
Known address filter patterns	12+ invalid patterns
State coverage (parser)	CA, NV (50 known cities pre-compiled)

This audit was generated from static analysis of the codebase including TypeScript types (4,353 lines), edge functions (process-upload, generate-insights, bulk-rescore), SQL migrations (229 files), and service layer code. For live data counts and actual violation type distribution, connect directly to the Supabase database and run analytical queries.